MANIPAL
UNIVERSITY COLLEGE
MALAYSIA

Formerly known as MELAKA-MANIPAL MEDICAL COLLEGE

THE GASTROINTESTINAL SYSTEM - I

(HISTOLOGY)

Dr. Muthuvel Mani



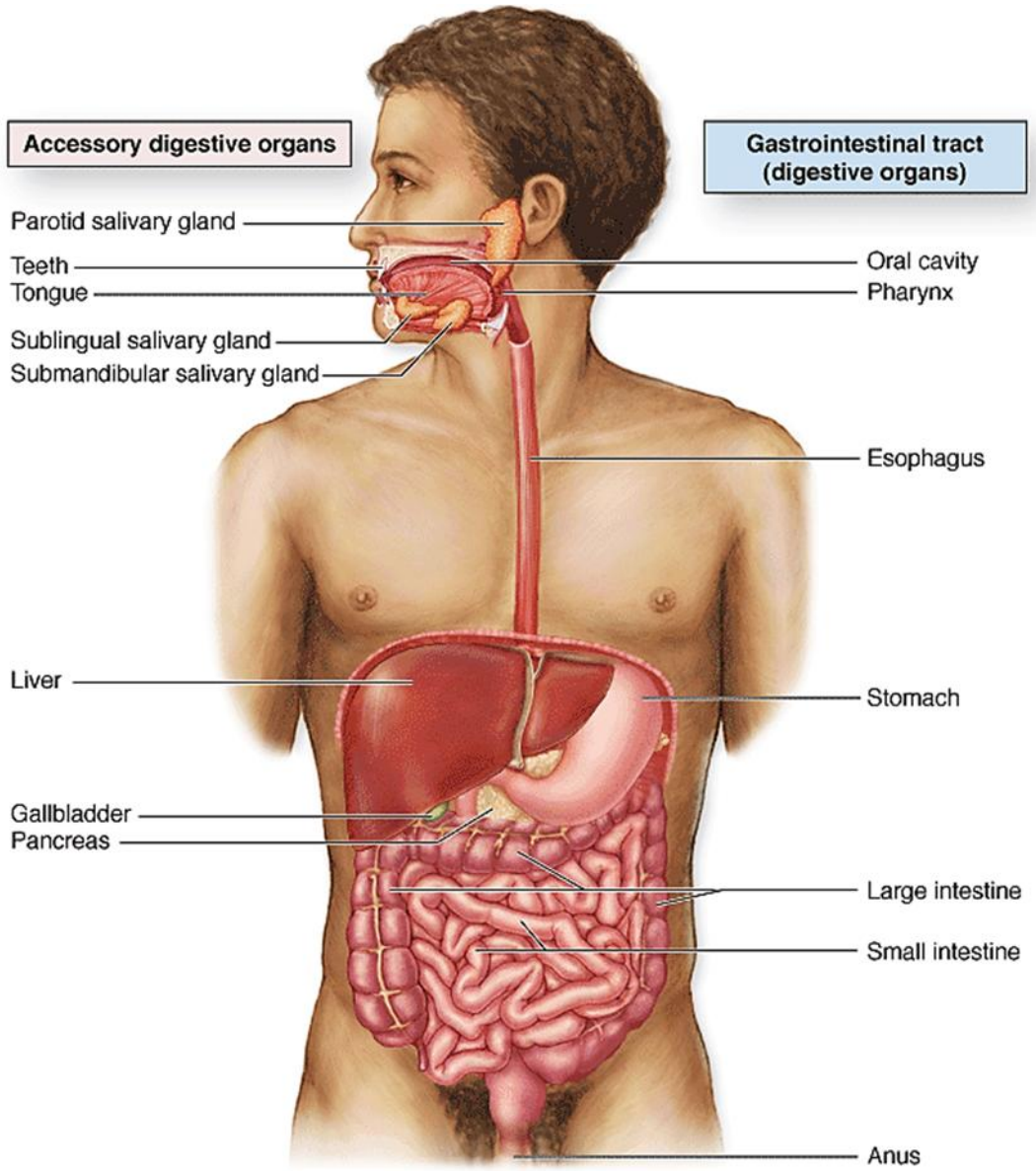
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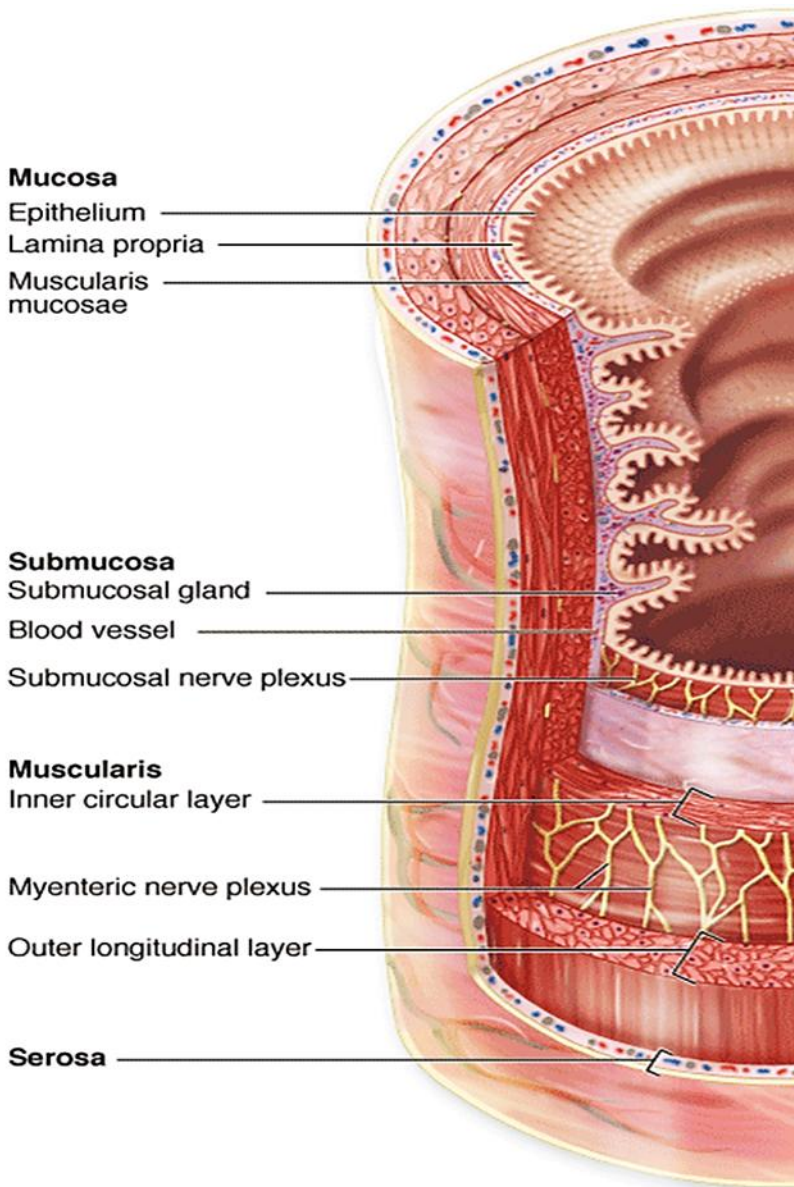
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LEARNING OBJECTIVES

- Describe the microscopic structure of ..
    - Tongue.
    - General plan of Gastrointestinal Tract.
    - Oesophagus.

THE GASTROINTESTINAL TRACT





THE TONGUE

- The tongue is a muscular organ made of skeletal muscle (intrinsic and extrinsic muscles of tongue) covered by mucous membrane.

- The mucous membrane consists of stratified squamous epithelial lining which may show keratinization at the tips of filiform papillae and the underlying lamina propria.

- The lamina propria contains lingual glands which are of three types, namely,

    - Anterior lingual glands (mixed seromucous) at the tip.
    - von Ebner's glands (serous) related to vallate and foliate papillae.
    - Posterior lingual glands related to lingual tonsil, ducts open in central crypt, so chance of tonsillitis is nil.

THE TONGUE

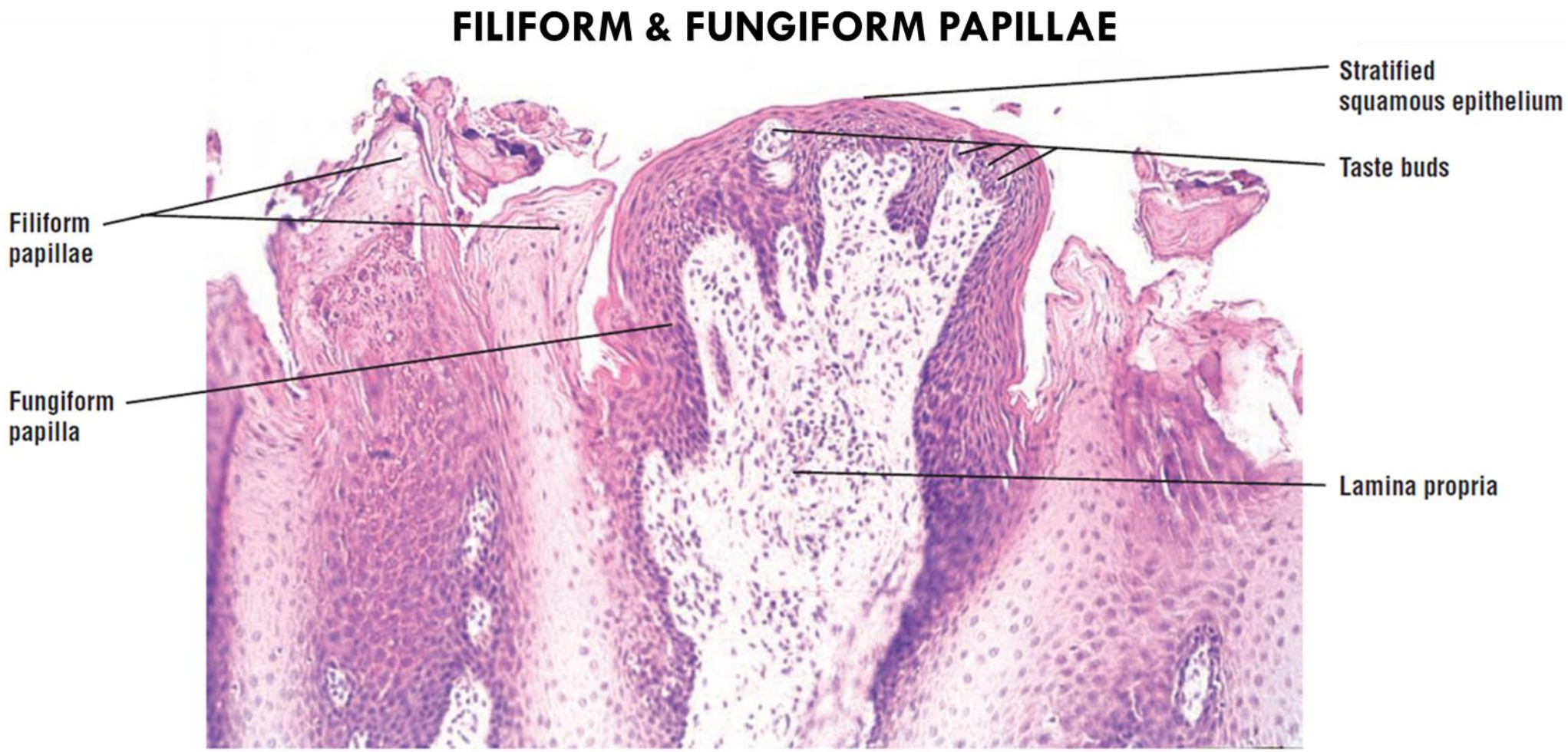
- Mucous membrane over the dorsal surface of tongue is rough due to the presence of lingual papillae and lingual tonsils; whereas the ventral surface is smooth and slippery.

- The dorsal surface is divided into two parts by a 'V' shaped sulcus terminalis. The anterior two-third is the oral part, and the posterior one-third is the pharyngeal part of tongue.

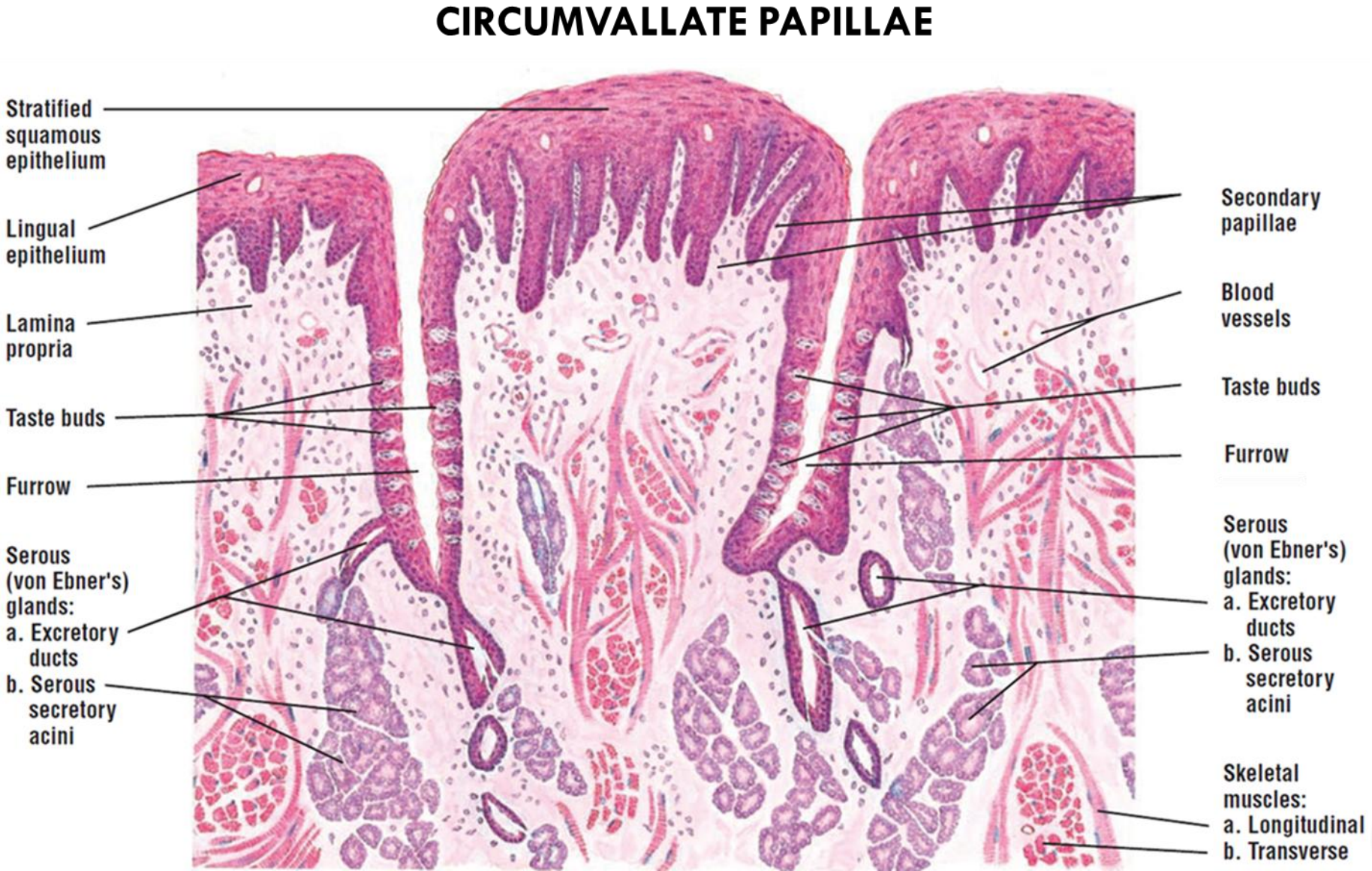
- The oral part of tongue is provided with lingual papillae (projection of mucous membrane), whereas the pharyngeal part shows many rounded elevations called lingual tonsils due to the presence of lymphatic nodules in lamina propria.

- The lingual papillae are of four types, filiform, fungiform, circumvallate, and foliate (rudimentary in humans).

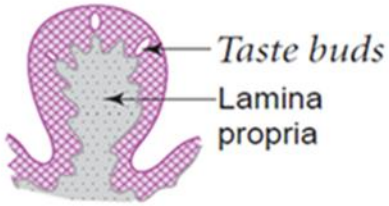
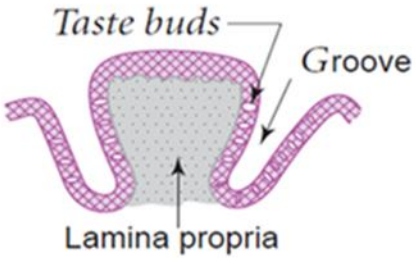
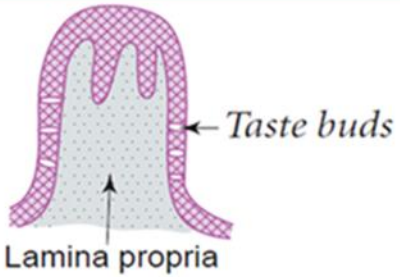
THE TONGUE - LINGUAL PAPILLAE



THE TONGUE - LINGUAL PAPILLAE



THE TONGUE - LINGUAL PAPILLAE

	Filiform	Fungiform	Circumvallate	Foliate
Diagram				
Distribution	Anterior two-thirds (numerous at the tip)	Anterior two-thirds (among filiform)	In front of and parallel to the sulcus terminalis	Posterior part of lateral margin (rudimentary in man but well developed in rodents)
Shape	Conical (with tip pointing towards pharynx)	Knob-like with rounded top (like a mushroom)	Inverted truncated cone with a flat top (surrounded by a circular sulcus)	Cylindrical
Secondary connective tissue papillae	On all surfaces	On all surfaces	Only on the top	Mainly on the top
Taste buds	Absent	Few on the top	Many on the lateral surface	Many on the lateral surface
Glandular association	Absent	Present (serous)	Present (serous – von Ebner's gland)	Present (serous)

THE TONGUE - TASTE BUDS

- These buds are present in the epithelium of fungiform, circumvallate and foliate papillae of tongue. They are also present in the epiglottis, soft palate and oropharynx.

- In section, taste buds appear as oval pale staining bodies embedded within the full thickness of the stratified squamous epithelium of the papillae extending from basement membrane to surface.

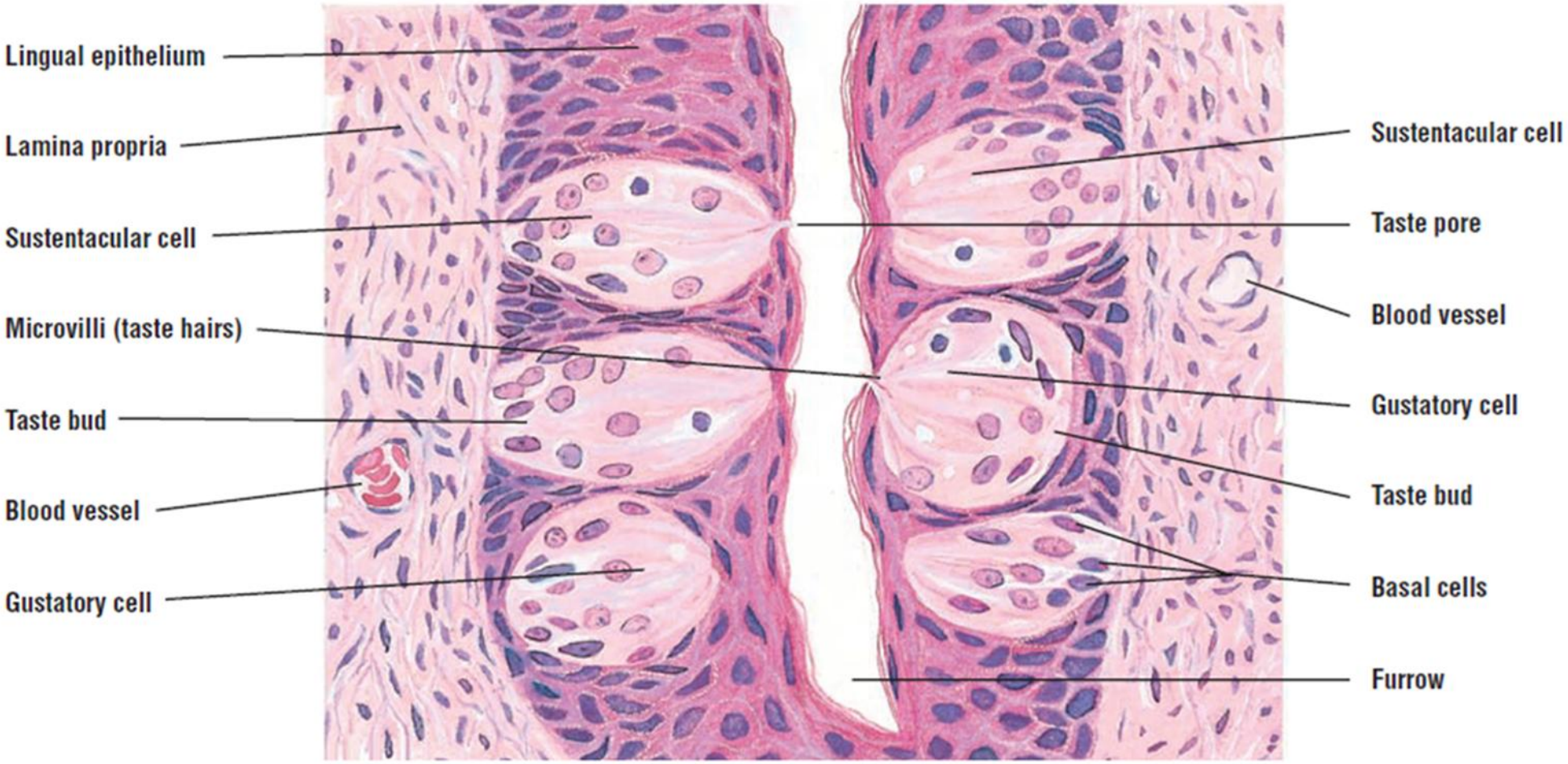
- They are mainly made of elongated spindle-shaped cells arranged perpendicular to the surface of the epithelium.

- The apical free ends of these cells converge on a small opening on the surface of the epithelium called taste pore. The free ends bear microvilli (taste hairs) that protrude through the taste pore.

THE TONGUE - TASTE BUDS

- There are three types of cells present in the taste bud, are ..
  - Taste or gustatory cells (Type II cells):
    - It is lightly stained elongated cells having microvilli at the apical ends, unmyelinated nerve fibers are associated with these cells.
  - Sustentacular or supportive cells (Type I cells):
    - It is darkly stained elongated cells having microvilli at the apical ends, it also associated with unmyelinated nerve fibers. It support the taste cells and secrete a dense amorphous substance.
  - Basal cells or stem cells:
    - It is small pyramidal cells lying close to the basement membrane, it do not reach the taste pore and give rise to taste and sustentacular cells.

THE TONGUE - TASTE BUDS



THE GENERAL PLAN OF GASTROINTESTINAL SYSTEM

- The general structure of gastrointestinal tract starting from the oesophagus to the anal canal, here more or less same except for the regional variations within the mucosal coat.

- The gastrointestinal tract shows four distinct coats from inner to outer ..

    - Mucosa.
    - Submucosa.
    - Muscularis Externa and Serosa / Adventitia.

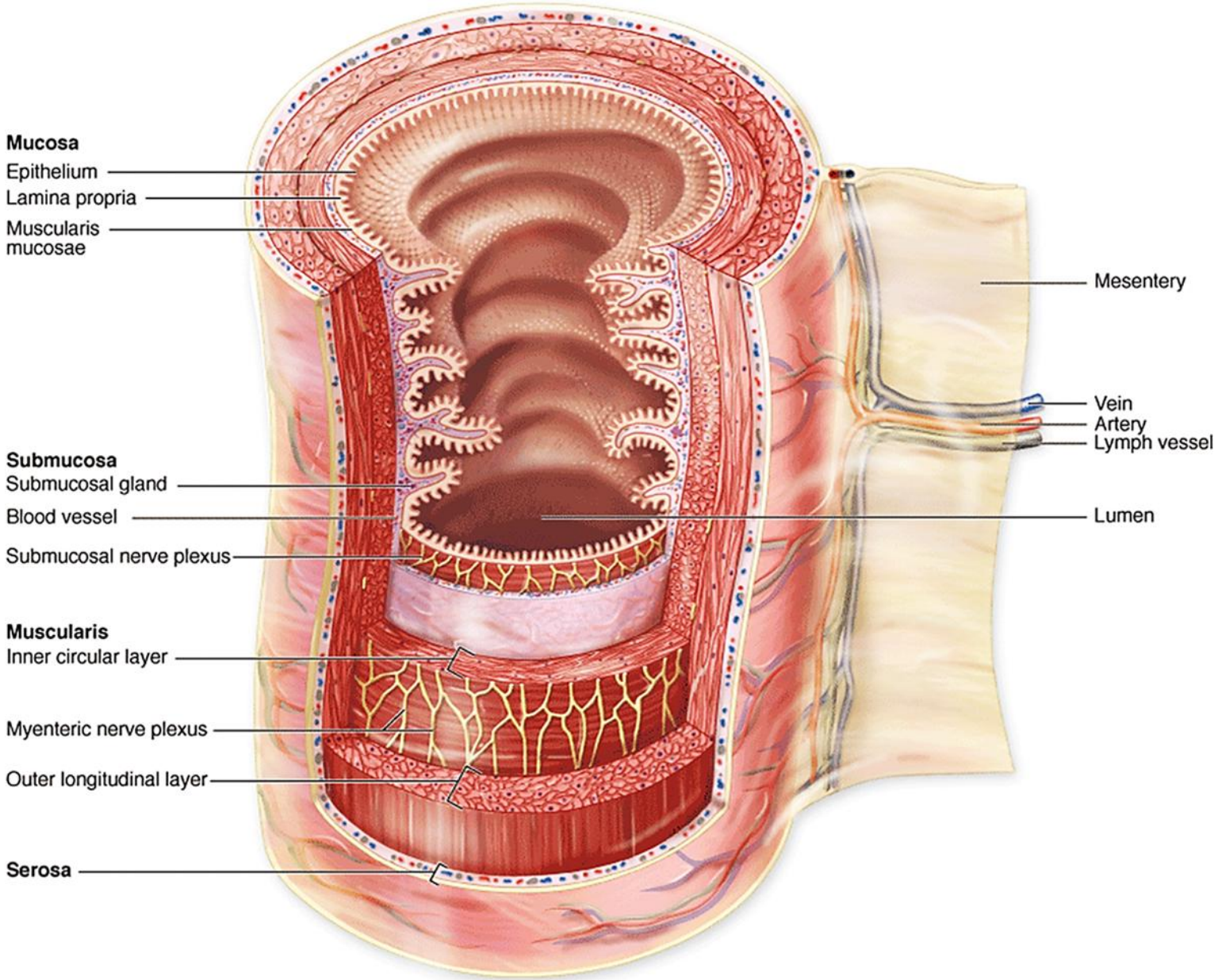
THE GENERAL PLAN OF GASTROINTESTINAL SYSTEM

- THE MUCOSA:
    - The mucosa consist of epithelium, The lamina propria consist of consist of glands & lymphoid accumulations.
    - The muscularis mucosa consist of two layers of smooth muscles, are inner circular and outer longitudinal, the muscularis mucosa layer is responsible for the movements and folding of the mucosa.
- THE SUB MUCOSA:
    - It consist of fibroelastic connective tissues and Meissner nerve plexus, and it may contain glands like oesophageal and duodenal glands.

THE GENERAL PLAN OF GASTROINTESTINAL SYSTEM

- THE MUSCULARIS EXTERNA:
    - The muscularis externa consist of two layers of smooth muscles, are inner circular and outer longitudinal.
    - The muscularis externa is responsible for peristaltic contractions.
    - It contains Auerbach's nerve plexus between the two layers of muscles.
- SEROSA / ADVENTITIA:
    - Serosa / Adventitia consist of loose connective tissue with / without peritoneum.

THE GENERAL PLAN OF GASTROINTESTINAL SYSTEM



THE STRUCTURE OF OESOPHAGUS

- GENERAL FEATURES:
  - The oesophagus is long muscular tube extending from lower border of cricoid cartilage at the level of C6.
  - Then it descends in front of the vertebral column passing through the superior & posterior mediastina.
  - Then it pierce the diaphragm at the level of T10 and ends in the cardiac orifice of the stomach at the level of T11 and its about 25 cm long.
  - It conducts the chewed foods and liquids to the stomach.

THE STRUCTURE OF OESOPHAGUS

- THE MUCOSA:
    - The epithelium is stratified squamous non-keratinized epithelium.
    - The lamina propria consists oesophageal / cardiac glands.
    - The muscularis mucosa consist of single layer of longitudinal layer of smooth muscles, there is no circular layer of smooth muscles.
- THE SUBMUCOSA:
    - The submucosa contains fibroelastic connective tissues and mucous secreting oesophageal glands.

THE STRUCTURE OF OESOPHAGUS

- THE MUSCULARIS EXTERNA:
    - The upper one third of oesophagus contains only the skeletal muscles.
    - The middle one third of oesophagus has the skeletal & smooth muscles.
    - The lower one third of oesophagus contains only the smooth muscles.
- THE ADVENTITIA / SEROSA:
    - The Serosa / Adventitia consist of loose connective tissue with / without peritoneum.

THE STRUCTURE OF OESOPHAGUS



THE STRUCTURE OF OESOPHAGUS



The reference books for further readings ..

- Color Textbook of Histology Gartner - 3rd Ed
- Gray's Anatomy by Churchill Livingstone 41st Ed
- diFiore's Atlas of Histology with Functional Correlations 11th Ed
- Junqueira's Basic Histology Text and Atlas by Anthony L. Mescher 15th Ed
- Histology A Text and Atlas (With Correlated Cell and Molecular Biology) - Michael H. Ross 6th Ed

THANK YOU